

Sunny Zhang

Toronto, ON, Canada | ssunny.zhang@mail.utoronto.ca | LinkedIn | GitHub | Portfolio

Education

University of Toronto, St. George

Honours BSc, Double Major in Computer Science and Mathematics

Expected 2029

GPA: 3.7/4.0

Technical Skills

Languages: Python, C++, TypeScript, SQL

Machine Learning and Data: PyTorch, Gymnasium, Stable-Baselines3, CleanRL, Pandas, NumPy

Frameworks: FastAPI, Next.js, Flask, D3

Tools and Platforms: Git, Docker, Supabase, Auth0, Hugging Face, pytest, Vercel

Experience

LLM/ML Research Intern

Jun 2026 – Present

Algoverse AI Research Program

- Investigating whether hierarchical rank pressure causes pretrained LLM trading agents to misreport executions or withhold profit-improving information, using simulator-derived ground truth rather than LLM judges.
- Designing an ablation study across situational features, observable logs, private trader chat, and private reasoning; encoding each source as a fixed feature block and training regularized logistic-regression detectors to isolate the predictive value of private reasoning.
- Evaluating next-round misconduct prediction using AUPRC, AUROC, calibration, and recall under a fixed inspection budget, with run-level train/test splits and held-out pressure settings to measure out-of-distribution generalization.

Projects

Tiny Society AI | Demo · Code · Research · Methods

Next.js, FastAPI, Python, Supabase, D3

- Built a full-stack research platform simulating how LLM agent societies form relationships, spread information, and shift beliefs over runs of up to 60 simulated days.
- Modeled agent memory, reflection, planning, and perception with explicit belief-update and influence equations, keeping behavior internally consistent across long-running simulations.
- Quantified emergent behavior through belief distribution, fragmentation, and influence metrics; authored two technical reports documenting the experimental design, findings, methods, and equations.
- Engineered a FastAPI simulation backend with bounded 6-call LLM concurrency, deterministic state application, and live Server-Sent Events streaming.

LaunchPilot | Devpost · GitHub

Next.js, TypeScript, FastAPI, Auth0, Backboard.io

- Built an AI launch copilot in 72 hours with a 2-person team at Hack Canada 2026; won the sole Backboard.io Best Use prize (1 of 59 teams) and placed top 3 of 94 in the SPUR startup track.
- Designed a supervised multi-agent workflow converting project briefs or GitHub repositories into market research, ICPs, positioning, messaging, KPIs, execution plans, and outreach drafts.
- Orchestrated three 5-agent stages with persistent threads and memory snapshots, gated by GitHub context, Auth0-scoped permissions, and human approval before execution.

UofT Theft Data Map | Live Demo · GitHub

Next.js, TypeScript, FastAPI, SQLite

- Built a full-stack app mapping Toronto Police theft-over-\$5,000 incidents around UofT, backed by a FastAPI pipeline that filtered campus-area records and normalized them in SQLite.
- Delivered an interactive OpenStreetMap heatmap with hotspot detection and summary metrics.